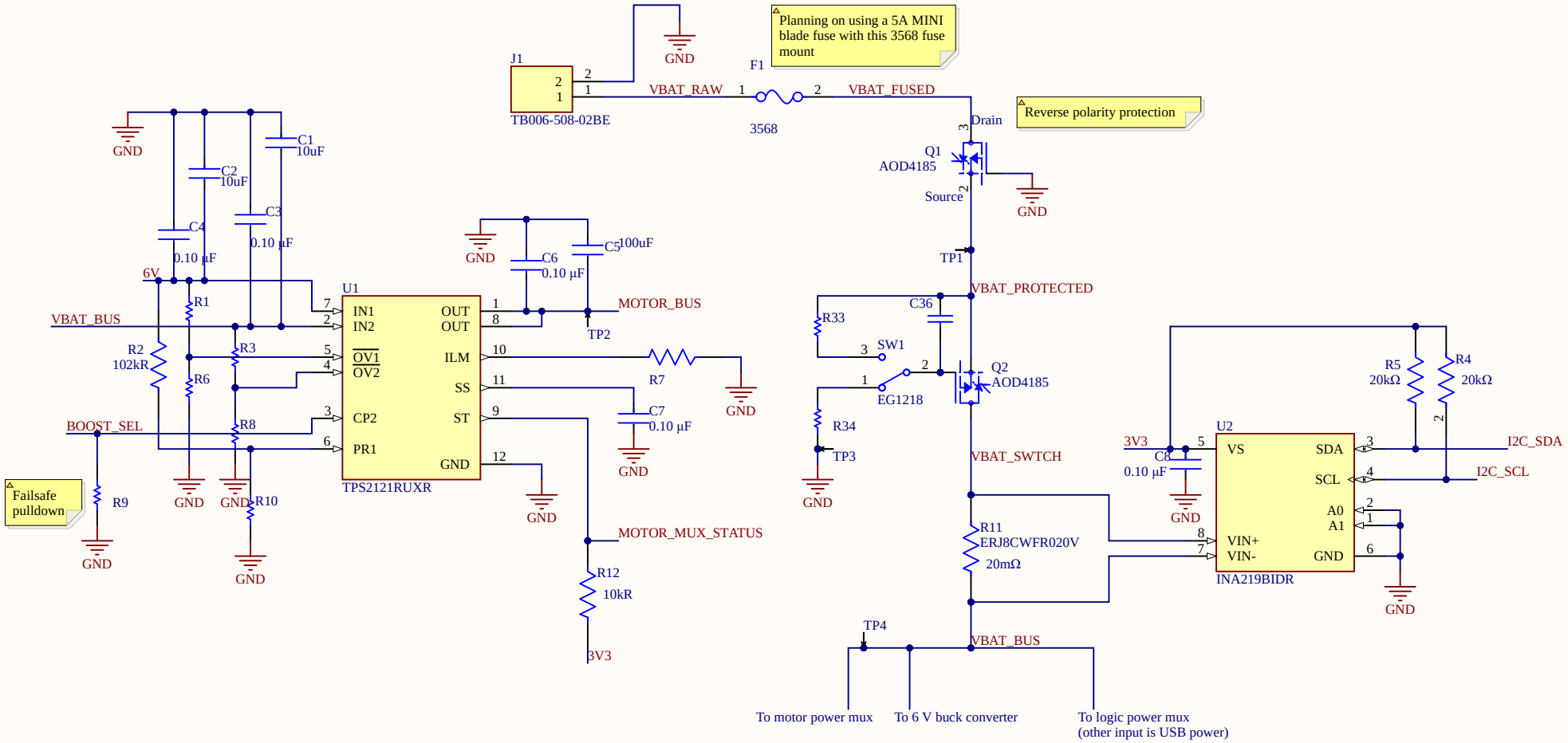
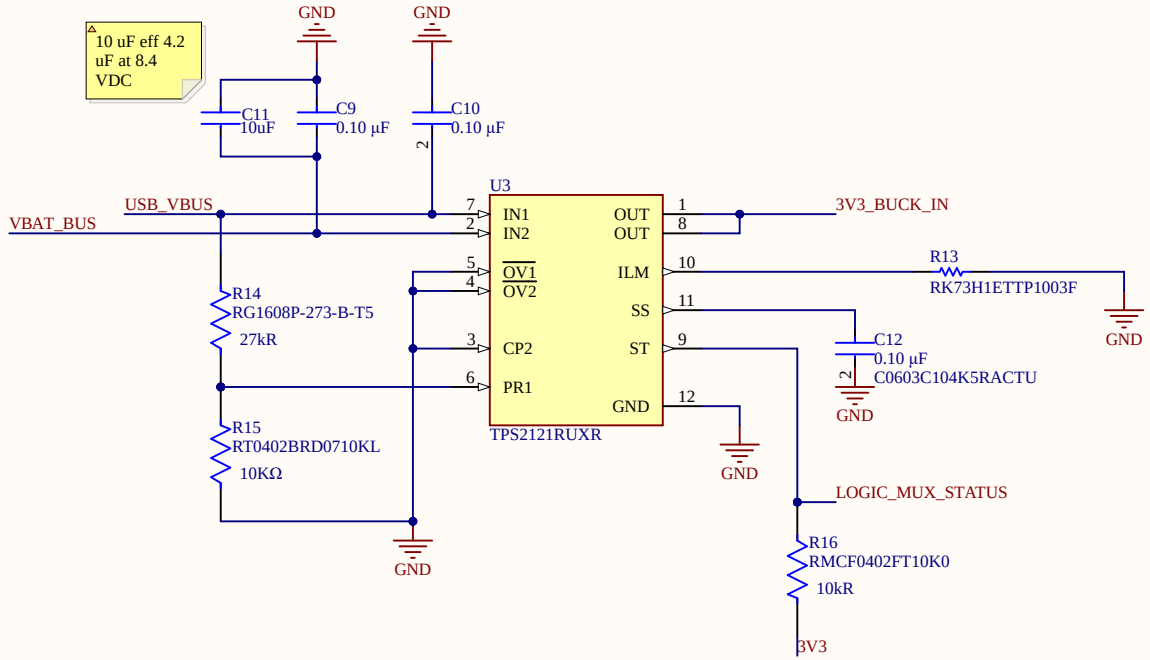
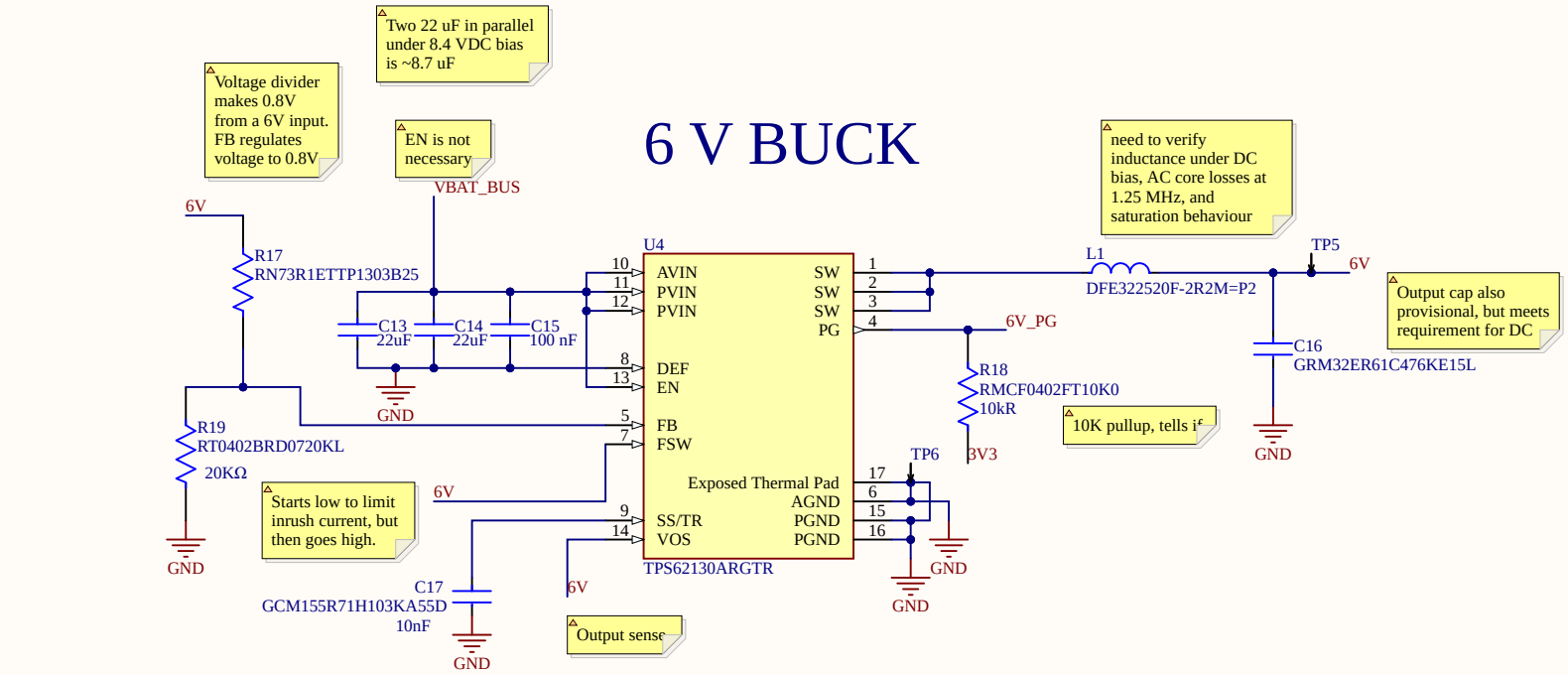
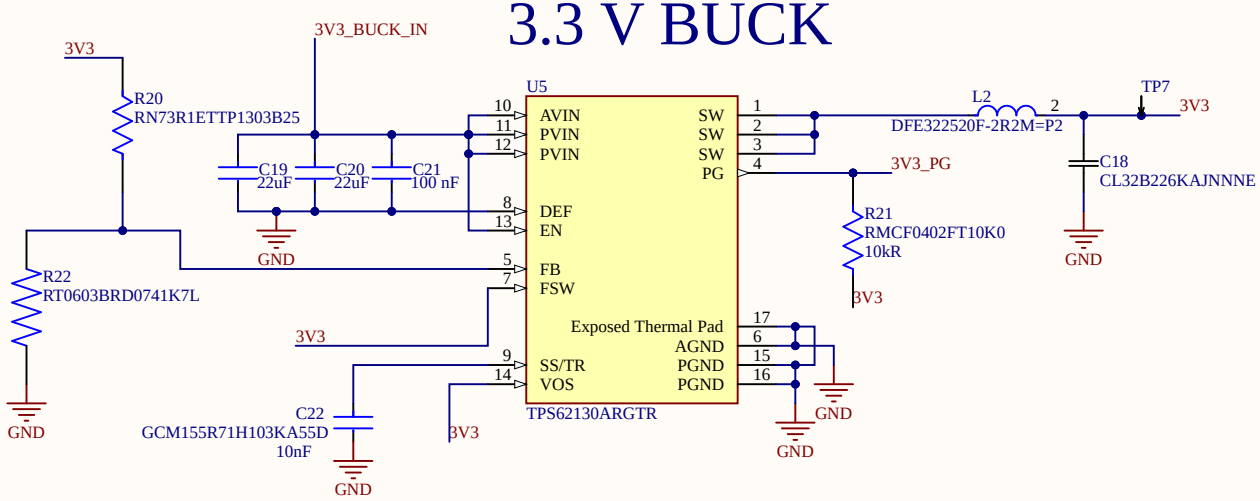


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File: C:\Users\..\Power.SchDoc		Drawn By: Marco Volpini

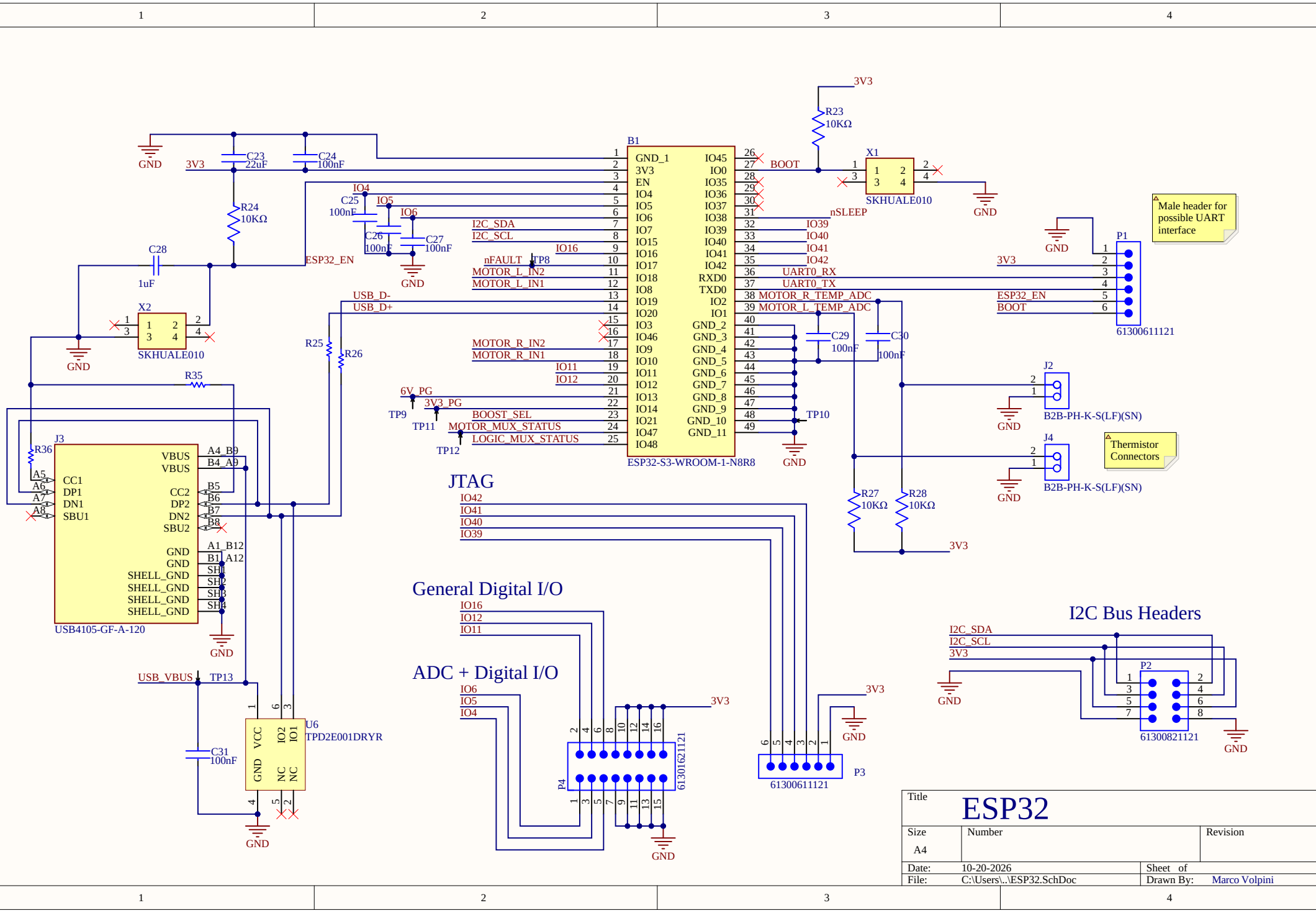


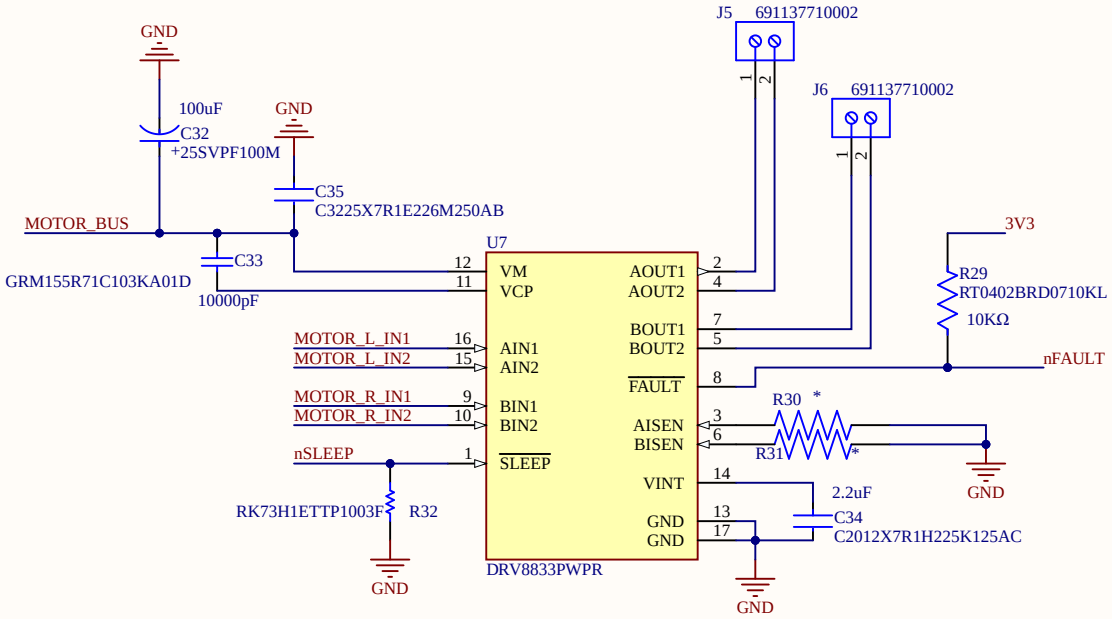
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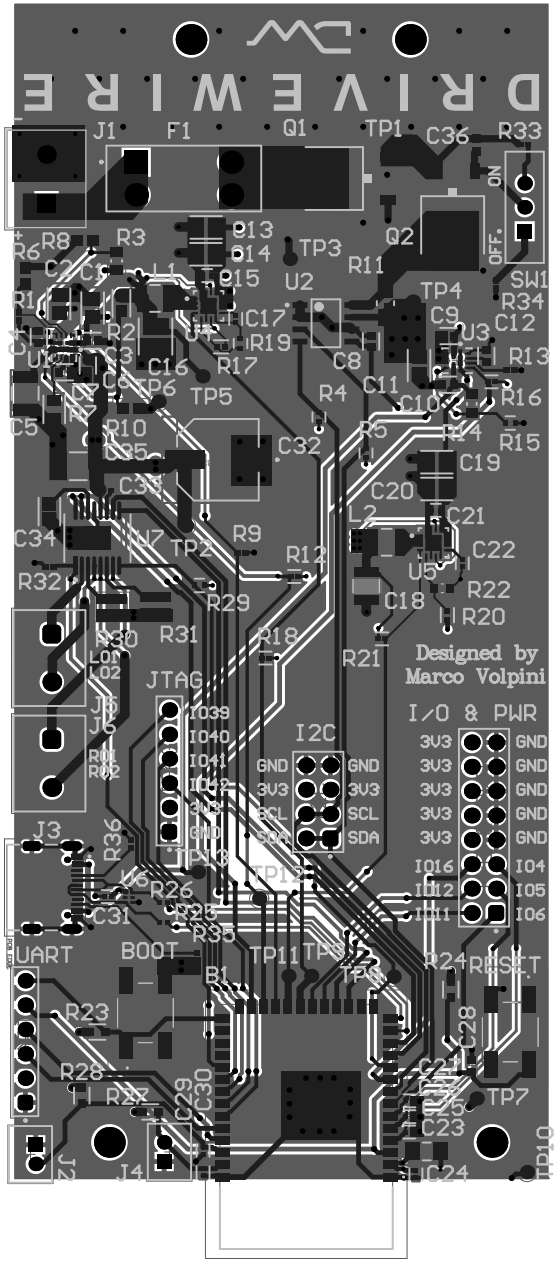


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Size: A4	Number:	Revision:	
Date: 10-20-2026		Sheet of	
File: C:\Users\..\buck_converters.SchDoc		Drawn By: Marco Volpini	


B1
ESP32-S3-WROOM-1-N8R8
GND_1
3V3
EN
IO4
IO5
IO6
IO7
IO15
IO16
IO17
IO18
IO8
IO19
IO20
IO3
IO46
IO9
IO10
IO11
IO12
IO13
IO14
IO21
IO47
IO48
IO45
IO0
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
RXD0
TXD0
IO2
IO1
GND_2
GND_3
GND_4
GND_5
GND_6
GND_7
GND_8
GND_9
GND_10
GND_11
3V3
R23
10KΩ
BOOT
X1
SKHUALE010
nSLEEP
IO39
IO40
IO41
IO42
UART0_RX
UART0_TX
MOTOR_R_TEMP_ADC
MOTOR_L_TEMP_ADC
C29
100nF
C30
100nF
R27
10KΩ
R28
10KΩ
P1
61300611121
Male header for possible UART interface
J2
B2B-PH-K-S(LF)(SN)
J4
B2B-PH-K-S(LF)(SN)
Thermistor Connectors
C23
22uF
C24
100nF
R24
10KΩ
C28
1uF
ESP32_EN
C25
100nF
C26
100nF
C27
100nF
I2C_SDA
I2C_SCL
nFAULT
TP8
MOTOR_L_IN2
MOTOR_L_IN1
USB_D-
USB_D+
MOTOR_R_IN2
MOTOR_R_IN1
6V_PG
3V3_PG
BOOST_SEL
MOTOR_MUX_STATUS
LOGIC_MUX_STATUS
TP9
TP10
TP11
TP12
X2
SKHUALE010
R25
R26
R35
R36
J3
USB4105-GF-A-120
VBUS
CC1
DP1
DN1
SBU1
CC2
DP2
DN2
SBU2
GND
SHELL_GND
A4_B9
B5
B6
B7
B8
A1_B12
B1_A12
USB_VBUS
TP13
U6
TPD2E001DRYR
VCC
GND
IO2
IO1
NC
C31
100nF
JTAG
General Digital I/O
ADC + Digital I/O
P4
61301621121
P3
61300611121
I2C Bus Headers
P2
61300821121
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ESP32
Size
A4
Number
Revision
Date:
10-20-2026
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File:
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Drawn By:
Marco Volpini



Title	Motor Driver	
Size: A4	Number:	Revision:
Date: 10-20-2026		Sheet of
File: C:\Users\..\motor_driver.SchDoc		Drawn By: Marco Volpini


DRIVEWIRE
Designed by
Marco Volpini
I/O & PWR
JTAG
I2C
UART
BOOT
RESET
ON
OFF

Level	Name	Description	Designator	Revision ID	Revision Note	Revision Status	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
	ISP52-53-WROOM-1-N8R8	WiFi Modules - 802.11 (Engineering Samples Only) ESP32-S2FH4 with 4 MB Octal-SPIRAM, 8 MB Quad SPI Flash, PCB Antenna	B1		Not managed		1				Mouser	356-ISP52WROOM1N8R8		
	GRM188R61H105KE19D	Chip Multilayer Ceramic Capacitors for General Purpose, 0603, 1Volt, X5R, 10% 10% 50V	C1, C2, C11	CMP-06035-004402-1	New From Design	Up to date	3	Murata	GRM188R61H105KE19L	Volume:Production				
	C0603C104K5RACTU	CAP CER 0.1UF 50V X7R 0603	C3, C4, C5, C7, C9, C10, C12, C13	CMP-0066-03061-2	New From Design	Up to date	8	Yageo Group	C0603C104K5RACTU	Volume:Production	Mouser	80-C0603C104K5R	0.07	0.7
	AMK063ABD475MH-F	CAP CAP 4.7UF 10V X5R 0603	C5	CMP-2000-05020-2	New From Design	Up to date	1	TAIYO YUDEN	AMK063ABD475MH-T	Obsolete	Mouser	963-EMK063ABD475MH-T	2.57	2.57
	CL21A226KOQNNNE	22uF 10Volts 20% 20 V Ceramic 0805 22uF 10Volts 10% Tolerance X5R Multilayer Ceramic Chip Capacitor	C13, C14, C18, C20	CMP-13271-000211-1	New From Design	Up to date	4	Samsung Electro-Mechanics	CL21A226KOQNNNG	Not Recommended for New Design	Mouser	187-CL21A226KOQNNNG	0.42	1.68
	CC0402KRX7R7BB104	CAP CERS 0.1UF 10V X7R 0402	C15, C21	CMP-03422-000048-1	New From Design	Up to date	2	Yageo Group	CC0402KRX7R7BB104	Volume:Production	Newark	CC0402KRX7R7BB104	0.009	0.09
	GRM32ER61C476KE15L	CAP CER 47UF 16V X5R 1210	C16	CMP-2000-06362-2	Released	Up to date	1	Murata	GRM32ER61C476KE15L	New Design	Not Recommended for New Design	DigiKey	490-5539-6-ND	1.43	1.43
	GCM155R71H103KA55D	Ceramic Capacitor for Automotive 10nF ±10% 50VDC X7R 0402 Paper T/R	C17, C22	CMP-2008-03723-3	New From Design	Up to date	2	Murata	GCM155R71H103KA55D	Volume:Production	ArrowElectronics	GCM155R71H103KA55D	0.0145	0.145
	CL31B226KAHNNNE	Description	C19		Not managed		1				Volume:Electronics	CL31B226KAHNNNE		
	CL31B226KPHNNNE	Cap Ceramic 22uF 10V X7R ±10% 0805	C23	CMP-13271-003516-1	New From Design	Up to date	1	Samsung Electro-Mechanics	CL31B226KPHNNNE	Volume:Production	Mouser	187-CL31B226KPHNNNE	0.5	0.5
	CL05B104KP5NNNC	Cap Ceramic 0.1uF 10V X7R ±10% 0402 +125°C Paper T/R	C24, C25, C26, C27, C28, C30, C31	CMP-13271-001069-1	New From Design	Up to date	7	Samsung Electro-Mechanics	CL05B104KP5NNNC	Volume:Production	DigiKey	1276-1002-1-ND	0.025	0.35
	GRM188R61A105KE19D	1uF, +/- 10%, 10 V, X5 on 85 degC, 0603(1608 Metric), RoHS, Tape and Reel	C29	CMP-2000-06336-1	Released	Up to date	1	Murata	GRM188R61A105KE19D	Not Recommended for New Design	ArrowElectronics	GRM188R61A105KE19D	0.106	0.106
	2R5MF100M	Conductive Polymer Aluminum Solid Capacitors 2.5V 100uF ±20% 25V RADIAL (5x5)5x5.4mm	C32	CMP-05428-000915-1	New From Design	Up to date	1	Panasonic	2R5MF100M	Volume:Production	Arrow	2R5MF100M	2.42	2.42
	GRM155R71C103KA01D	Chip Capacitor, 10 nF, 16 V, 10%, X5 on 125 degC, 0402 (1005 Metric), RoHS, Tape and Reel	C33	CMP-2000-06332-1	Released	Up to date	1	Murata	GRM155R71C103KA01D	Not Recommended for New Design	ArrowElectronics	GRM155R71C103KA01D	0.0964	0.0964
	C3012X7R1H225K125AC	Multilayer Ceramic Capacitors 2.2uF ±10% 50V X7R SMD 0805	C34	CMP-2000-06040-2	Released	Up to date	1	TDK	C3012X7R1H225K125AC	Volume:Production	ArrowElectronics	C3012X7R1H225K125AC	0.4917	0.4917
	C3225X7R1E226M250AB	Multilayer Ceramic Capacitors 22uF ±20% 25V X7R SMD 1210	C35	CMP-06246-001769-2	New From Design	Up to date	1	TDK	C3225X7R1E226M250AB	Volume:Production	ArrowElectronics	C3225X7R1E226M250AB	1.26	1.26
	C0603C105K8PACTU		C36		Unknown owner		1							
	9268		J1	CMP-2000-05464-2	Released	Up to date	1	Keystone	9268	Volume:Production	Newark	9268	1.56	1.56
	TB006-508-03BE	2~36Poles, Screw Type, Horizontal, 5.08 Pitch, 26~14 AWG, Terminal Block Connector	J3		Not managed		1	Same Sky	TB006-508-03BE	Volume:Production	ArrowElectronics	TB006-508-03BE	0.1624	100.04
	B2B-PH-K-S(LF)(SN)	Male Header, Pitch 2mm, 1x2 Position, Height 5 mm, Tail Length 3.4 mm, -25 to 85 degC, Tape and Reel	J5, J6	CMP-2000-05162-1	Released	Up to date	2	JST	B2B-PH-K-S(LF)(SN)	Volume:Production	Newark	B2B-PH-K-S(LF)(SN)	0.099	0.198
	USB4105-GF-A-120	USB CONN, 2.0 TYPE-C, 6CPT, 16POS, SMT, USB Connector Type:USB Type-C, USB Standard:USB 2.0, Gender:Receptacle, No. of Positions:16Position, Connector Mounting:Surface Mount, Through Hole Mount, Orientation:Right Angle RoHS Compliant: Yes	J8		Not managed		1				Newark	USB4105-GF-A-120		
	691137710002	Terminal Block, 5mm Pitch, Horizontal, Pressure Clamp Screw Wire, Pcb, 2Pole, Gr	J9, J6	CMP-1502-03324-3	Released	Up to date	2	Wurth Elektronik	691137710002	Volume:Production	DigiKey	732-10955-ND	0.38	0.76
	DR322532F-282MHF2	DR322532F-282MHF2 Inductor Power Shielded Wirewound 2.8uH 10% 1MHz Metal Alloy 0.042Ohm 20%(1210) SMD	L1, L2	CMP-00260-1303816581-1	New From Design	Up to date	2	Murata	DR322532F-282MHF2	Volume:Production	ArrowElectronics	DR322532F-282MHF2	0.2692	0.5384
	61300611121	90°Vertical Pin Header WR-PHD, Pitch:2.54 mm, Single Row, Eqing	P1, P3	CMP-1502-01070-4	Released	Up to date	2	Wurth Elektronik	61300611121	Volume:Production	DigiKey	732-5319-ND	0.33	0.66
	61300821121	90°Vertical Pin Header WR-PHD, Pitch:2.54 mm, Dual Row, Eqing	P2	CMP-1502-01078-4	Released	Up to date	1	Wurth Elektronik	61300821121	Volume:Production	DigiKey	732-5296-ND	0.67	0.67
	61301621121	90°Vertical Pin Header WR-PHD, Pitch:2.54 mm, Dual Row, 16 pins	P4	CMP-1502-01097-4	Released	Up to date	1	Wurth Elektronik	61301621121	Volume:Production	DigiKey	732-5299-ND	1	1
	ADI4185	P-Channel 40V 60A TO-219AB/TO-220/TO-262/TO-263/TO-220-Tab	Q1, Q2		Not managed		2				DigiKey	785-1222-1-ND		
	RK73H1ETP5011F		R1	CMP-2002-02895-1	Released	Up to date	1	KOA Speer	RK73H1ETP5011F	Volume:Production	DigiKey	2019-RK73H1ETP5011FCT-ND	0.1	0.1
	RMCF0603FG10K0	100kOhm 1% 1W 0603 Thick Film Chip Resistor AEC-Q200 Compliant	R3	CMP-20527-000764-1	New From Design	Up to date	1	Rackspace Electronics	RMCF0603FG10K	Volume:Production	DigiKey	738-RMCF0603FG10KCT-ND	0.1	0.1
	RC0805FR-0749R9L		R3	CMP-2001-01136-1	Released	Up to date	1	Yageo Group	RC0805FR-0749R9L	Volume:Production	ArrowElectronics	RC0805FR-0749R9L	0.044	0.044
	RC1402FR-0710KL		R4, R5	CMP-2002-07001-2	Released	Up to date	2	Yageo Group	RC1402FR-0710KL	Volume:Production	ArrowElectronics	RC1402FR-0710KL	0.0307	0.061
	TMPU0603M6R80KHX		R6, R8, R10	CMP-2000-03683-1	Released	Up to date	3	Vishay	TMPU0603M6R80KHX	Volume:Production	DigiKey	TMPU0R8M4CT-ND	2.1	6.3
	RK73R2ATTE2060BD	206kOhm 1% 0.125W, 1/8W Chip Resistor 0805 (2012 Metric) Moisture Resistant Thin Film	R7	CMP-04015-002765-1	New From Design	Up to date	1	KOA Speer	RK73R2ATTE2060BD	Volume:Production	DigiKey	2019-RK73R2ATTE2060BDCT-ND	0.13	0.13
	RK73H1ETP3003F		R9, R13, R20, R34	CMP-2002-04173-1	Released	Up to date	4	KOA Speer	RK73H1ETP3003F	Volume:Production	ArrowElectronics	RK73H1ETP3003F	0.0154	0.154
	ERJ8CWFR005V		R11	CMP-2003-00793-2	Released	Up to date	1	Panasonic	ERJ8CWFR005V	Volume:Production	ArrowElectronics	ERJ8CWFR005V	0.4242	0.4242
	RMCR0402FT10K0	10kOhm 1% 0.063W 0402 Thick Film Chip Resistor AEC-Q200 Compliant	R12, R24, R48, R51	CMP-20527-000139-1	New From Design	Up to date	4	Rackspace Electronics	RMCR0402FT10K0	Volume:Production	Mouser	708-RMCR0402FT10K0	0.012	0.12
	RG1608P-273-B-T5	Automotive Metal Film Rim Chip Resistor, 0603, 27kΩ, 0.1%, 25ppm/°C, 0.1W, 100V	R14	CMP-2000-02127-2	Released	Up to date	1	Susumu	RG1608P-273-B-T5	Volume:Production	ArrowElectronics	RG1608P-273-B-T5	0.0964	0.0964
	RT0402BRD0710KL	10kΩ 0.1% Orlel 1/16W 1% 0402	R15, R29	CMP-2002-02915-2	New From Design	Up to date	2	Yageo Group	RT0402BRD0710KL	Volume:Production	ArrowElectronics	RT0402BRD0710KL	0.0964	0.1928
	RK73R1ETP1302D	13kOhm 1% 0.063W, 1/16W Chip Resistor 0402 (1005 Metric) Moisture Resistant Thin Film	R17, R20	CMP-04015-027314-1	New From Design	Up to date	2	KOA Speer	RK73R1ETP1302D	Volume:Production	DigiKey	2019-RK73R1ETP1302DSCT-ND	0.1	0.2
	RT0402BRD0720KL	10 Ω 0.1% 1/16W 1% 0402	R19	CMP-2002-02936-2	New From Design	Up to date	1	Yageo Group	RT0402BRD0720KL	Volume:Production	ArrowElectronics	RT0402BRD0720KL	0.0964	0.0964
	RT0603BRD0741KL	51.1 kOhms 1% 1/10W 0603 Thin Film	R22		Not managed		1				Mouser	603-RT0603BRD0741KTL		
	RT0603BRD0710KL	10kΩ 0.1% Orlel 1/10W 1% 0603 (1608 Metric) Thin Film	R23, R24, R27, R28	CMP-03412-000619-1	New From Design	Up to date	4	Yageo Group	RT0603BRD0710KL	Volume:Production	ArrowElectronics	RT0603BRD0710KL	0.0964	0.3856
	CRCW0402220RFKED		R25, R26	CMP-2002-09382-1	Released	Up to date	2	Vishay	CRCW0402220RFKED	Volume:Production	ArrowElectronics	CRCW0402220RFKED	0.0024	0.0048
	ERJ8CYR51V		R30, R31		Not managed		0				Newark	ERJ8CYR51V		
	RK73H1ETP3032F		R32		Unknown owner		1							
	RT0402FRE07154KL		R35, R36		Unknown owner		2							
	EG2218	Slide Switch, SMD, -30 to 70 degC, 3 Pin THD, Rbt=5	SW1	CMP-1696-00002-1	Released	Up to date	1	C-S Switch	EG2218	Volume:Production	DigiKey	EG1903-ND	0.72	0.72
	TestPoint		TP1, TP2, TP3, TP4, TP5, TP6, TP7, TP8, TP9, TP10, TP11, TP12, TP13		Not managed		13							
	TPS2113ALHR	POWER MGMT SWITCH/VMGMT	U1, U3	CMP-04910-000009-1	New From Design	Up to date	2	Texas Instruments	TPS2113ALHR	Volume:Production	DigiKey	296-53410-6-ND	2.44	4.88
	RAQ108BOR	IC SWITCH PWRCLKRSBOR10 SOIC	U2	CMP-07704-000005-3	Released	Up to date	1	Texas Instruments	RAQ108BOR	Volume:Production	ArrowElectronics	RAQ108BOR	1.88	1.88
	TPS62130ARGTR	IC REGULATOR ADJ 0.5V 3A 16QFN	U4, U5	CMP-04918-000067-1	New From Design	Up to date	2	Texas Instruments	TPS62130ARGTR	Volume:Production	Newark	TPS62130ARGTR	1.99	3.98
	TPS3830XDRR	IC SUPERVISOR 1 CH 10 WSON	U6	CMP-3326-00025-3	Released	Up to date	1	Texas Instruments	TPS3830XDRR	Volume:Production	ArrowElectronics	TPS3830XDRR	0.3021	0.3021
	SRA8833PAPR	IC MOTOR DRIVER 1.8A 16HTSSOP	U7	CMP-1529-00006-3	Released	Up to date	1	Texas Instruments	SRA8833PAPR	Volume:Production	Mouser	595-SRA8833PAPR	3.03	3.03
	SHHLALE010	Slide Switches 0.04A 0x2 5mm 180gf	X1, X2		Not managed		2	ALPS	SHHLALE010	Volume:Production	Newark	SHHLALE010	0.3873	0.87